Calibration Surface Analysis Across Price and Time Dimensions

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Prediction markets - generate continuous streams of probabilistic quotes structurally analogous to option prices. Drawing on the volatility-surface calibration literature and the time–space decoupling methodology of streak-tube imaging lidar (Chen et al., 2023), we introduce the prediction market calibration surface C(K, τ): a two-dimensional function over strike K ∈ [0,1] (probability threshold) and remaining horizon τ (days to resolution). We formalise calibration error as a structured surface, develop a two-stage decomposition separating the price skew CK(K) from the temporal drift Cτ(τ), introduce the Market Coherence Index (MCI), and show that surface flattening - a well-known warning signal in options markets - has a direct analogue in prediction market microstructure.

Introduction

The classical view of prediction market calibration is scalar: aggregate quoted probabilities into bins, compare bin averages against empirical frequencies, and assess whether the market "knows what it does not know." If events assigned 0.70 resolve positively 70% of the time, the market passes the test.

This view is incomplete. It conflates two fundamentally different sources of systematic bias. The first is a price-axis distortion: the market may systematically overestimate extreme probabilities and underestimate mid-range ones - analogous to the implied-volatility smile in options. The second is a time-axis distortion: the market may drift predictably as an event approaches, undershooting near-certain outcomes in final days or overshooting long-horizon events due to narrative entrenchment.

Recent commentary on volatility surface calibration (Browne, 2024) warns that when the vol surface flattens - loses its characteristic smile and term-structure curvature - arbitrage coherence has broken down. We argue that an analogous flattening phenomenon occurs in prediction markets and is equally diagnostic.

Core thesis: the calibration error of a prediction market is not a scalar constant but a structured surface with characteristic smiles, skews, and term-structure effects. Treating it as a point estimate discards the information needed to identify and exploit systematic mispricings.

Theoretical Framework

From Point Calibration to a Surface

The standard Brier Score provides an aggregate calibration metric. For N events with quoted probabilities $\hat{p}_i$ and binary outcomes $o_i \in \{0,1\}$:

$o_i \in \{0,1\}$ is the realised outcome; $\hat{p}_i$ is the market-quoted probability. The score is a single number — it has no memory of where in (K, τ) space the error originated.

We replace this with a conditional expectation. Define the calibration surface C(K, τ) as the expected signed deviation of market price from true probability, conditional on the current quote being near K and remaining horizon near τ:

$p^*(t)$ denotes the latent true probability; $\hat{p}(t)$ is the market quote; the expectation is over the empirical distribution of historical events. C > 0 means the market overprices; C < 0 means it underprices.

Nonparametric Surface Estimation

Estimating C(K, τ) from finite data requires a smoother. We adopt Nadaraya–Watson kernel regression with a bivariate Gaussian kernel:

$\delta_i = \hat{p}_i - o_i$ is the observed residual; $K_h(u,v) = \exp(-u^2/2h^2_K - v^2/2h^2_\tau)$ is a Gaussian kernel with bandwidths h_K and h_τ selected by cross-validation.

The Options Analogy

In options theory, the Black–Scholes model assumes a single implied volatility σ. The volatility smile - the empirical U-shape of σ(K,T) in K and downward slope in T - captures the market's correction of that assumption. Our C(K,τ) is the direct prediction-market analogue, encoding systematic patterns by which the market corrects its aggregate bias as a function of both probability level and time horizon.

The conditional standard deviation of calibration error — the prediction-market analogue of implied volatility. High IPV signals noisy, uncertain regions of the (K, τ) surface.

Two-Stage Decomposition

Following the time–space decoupling methodology of Chen et al. (2023) - who separated streak-tube imaging errors into a temporal calibration array and an angular calibration array - we decompose C(K, τ) additively:

C_K(K) is the price-axis skew (the "smile"); C_τ(τ) is temporal drift; C_int(K, τ) is the interaction term capturing non-separable structure — fit by sequential regression residuals.

Stage One - Temporal Calibration

Analogous to Chen et al.'s time calibration experiment - sweeping the laser delay step-by-step across the CCD to isolate the time-dimension offset - we bin events by horizon τ and estimate the marginal drift:

$\mathcal{S}_j = \{i : \tau_i \in [\tau_j - \Delta/2, \tau_j + \Delta/2]\}$; smoothed with a cubic spline $\hat{s}_\tau(\tau)$. This is the prediction-market equivalent of the time calibration array in lidar processing.

Stage Two - Price Calibration

After removing temporal structure, residuals $\delta_i^* = \delta_i - \hat{s}_\tau(\tau_i)$ are regressed on K, yielding the smile/skew analogous to the angular calibration array in Chen et al.:

A degree-4 polynomial fit to bin-averaged residuals δ_i^* as a function of K. Higher-order terms capture the asymmetric skew at extreme probability levels (K < 0.10 and K > 0.90).

No-Arbitrage Constraint

For any pair of complementary events A and Ā with the same resolution date, market consistency requires $\hat{p}(A) + \hat{p}(\bar{A}) = 1$. In terms of the calibration surface:

The surface must be antisymmetric about K = 0.5. Any violation indicates a measurable arbitrage: one side is systematically mispriced relative to its complement.

Empirical Visualisation

The Market Coherence Index

A single-number summary of calibration surface quality is useful for comparing platforms and tracking market health over time. The Market Coherence Index MCI(τ) is one minus the weighted absolute bias integrated over the K-axis:

w(K) = K(1−K) penalises mid-range errors more heavily than extreme-probability errors. MCI ∈ [0,1]; MCI = 1 is perfect calibration.

Surface flattening - the pathology identified in options markets by Browne (2024) - corresponds to C(K, τ) collapsing toward a constant. MCI drops

sharply in this regime because the numerator integral does not shrink to zero. Empirically, MCI values below 0.80 correlate with low liquidity and wide bid-ask spreads on major prediction platforms.

> When MCI drops sharply in the short-horizon region ($\tau < 7$ days), it is a reliable early warning that the market has lost its price-discovery function near resolution - analogous to volatility surface flattening before option expiry.

Anomaly Taxonomy

The two-stage decomposition reveals five recurring calibration anomaly patterns across major prediction market platforms:

> *"Surface flattening is not the absence of information - it is the signal that the market no longer knows where to look for it"*

Practical Applications

Calibration-Adjusted Probability
Given the estimated surface $\hat{C}(K, \tau)$, a participant can compute a corrected probability that removes systematic bias. The point estimate and its uncertainty are:

Delta-Neutral Portfolio Construction
Analogously to delta-hedging in options, a trader wishing to exploit the price skew $C_K(K)$ without exposure to the temporal drift $C_\tau(\tau)$ can construct a portfolio of events that is τ-neutral: positions are balanced across horizons so that aggregate C_τ exposure nets to zero, isolating the smile opportunity. This is the prediction-market equivalent of a calendar-spread arbitrage in options.

Platform Quality Benchmarking

The MCI(τ) curve, computed over a rolling window of resolved events, provides a time-varying quality signal for any platform. A market with MCI > 0.88 across all $\tau \in [7, 365]$ is a superior information aggregator compared to one that achieves high MCI only in the comfortable mid-range. Regulators and researchers can use MCI as a transparency metric analogous to the bid-ask spread in financial markets.

Conclusions

We have proposed a formal framework for analysing prediction market calibration through the lens of a two-dimensional surface C(K, τ). The principal contributions are as follows.

First, the scalar Brier Score systematically conceals structured biases visible only under the two-dimensional view. Second, the decomposition C = C_K + C_τ + C_int - inspired by the time–space decoupling approach in streak-tube imaging lidar - separates price skew from temporal drift, enabling targeted correction of each. Third, the Market Coherence Index MCI(τ) is a sensitive indicator of loss of market information structure, providing the prediction-market analogue of volatility-surface flattening. Fourth, the no-arbitrage antisymmetry condition C(K,τ) = −C(1−K,τ) yields a testable constraint that, when violated, signals an exploitable mispricing.

A natural extension is to the three-dimensional surface C(K, τ, V) incorporating trading volume V - analogous to order-flow skew in volatility surface models - which preliminary evidence suggests is substantial at the short-horizon wing.

Reference List

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